Customer Features & Signals

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1. Customer Features & Signals

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a. Spend Decline Flag

Definition:

Tracks if a customer's spending has declined significantly over a given period.

Logic (Description in Detail):

This feature compares a customer's current spend to their historical spending patterns to detect if there's a decline. By tracking this over a defined period (e.g., 30 days, 60 days), we can identify customers who are spending less, flagging them for potential recovery actions.

Formula:

Spend decline percentage = ((Previous Spend – Current Spend) / Previous Spend) × 100

Variable Values:

- **Previous Spend (PS)**: Total spend in the past 30 days (e.g., ₹15,000).
- **Current Spend (CS)**: Total spend in the current 30 days (e.g., ₹10,000).

Threshold:

- **Decline Threshold**: flag if Spend Decline Percentage exceeds 20% over 30 days.

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b. Days Since Last Purchase

Definition:

Calculates the number of days since the customer last made a purchase.

Logic (Description in Detail):

This feature measures customer inactivity by tracking how many days have passed since their last order. If this value exceeds a certain threshold (e.g., 30, 60, 90 days), it signals the customer as inactive or dormant.

Formula:

Days Since Last Purchase = Current Date - Last Purchase Date

Variable Values:

- **Current Date (CD)**: The current date (e.g., "2026-04-09").
- **Last Purchase Date (LPD)**: Date of the last purchase (e.g., "2026-03-10").

Threshold:

- **Inactive Threshold**: 30 days → Flagged if Days Since Last Purchase > 30.

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c. Purchase Frequency

Definition:

Measures how frequently a customer makes a purchase within a given period.

Logic (Description in Detail):

Purchase frequency shows how many times a customer has bought products in a defined time window. A decrease in frequency can indicate that the customer is losing interest, so this feature tracks and flags customers with declining purchase frequency.

Formula:

Purchase frequency = Total purchases ÷ Time window (e.g., 30 days)

Variable Values:

- **Total Purchases (TP)**: Total number of purchases in the time window (e.g., 5 purchases).
- **Time Window (TW)**: Defined time period (e.g., 30 days).

Threshold:

- **Low Frequency Threshold**: Below 1 purchase per week → Flagged if Purchase Frequency < 1.

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d. Average Order Value (AOV)

Definition:

The average amount of money spent by a customer per order.

Logic (Description in Detail):

This feature tracks the average value of a customer's orders. A significant drop in AOV can indicate that customers are purchasing fewer high-value items, which can signal a decrease in customer engagement or interest.

Formula:

AOV = Total Spend ÷ Total Orders

Variable Values:

- **Total Spend (TS)**: Total amount spent by the customer (e.g., ₹5,000).
- **Total Orders (TO)**: Total number of orders placed by the customer (e.g., 10).

Threshold:

- **AOV Drop Threshold**: Flag if current AOV is less than 80% of previous AOV: Current AOV < Previous AOV × 0.8.

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e. Spend Velocity

Definition:

Measures how quickly a customer is spending over a defined period.

Logic (Description in Detail):

Spend velocity is calculated by comparing how quickly a customer is spending money compared to their previous spending pattern. A decrease in velocity means the customer is purchasing less frequently, which can signal disengagement.

Formula:

Spend velocity = Total spend ÷ Time between purchases (days)

Variable Values:

- **Total Spend (TS)**: Total amount spent in a defined period (e.g., ₹2,000).
- **Time Between Purchases (TBP)**: Average time interval between two purchases (e.g., 10 days).

Threshold:

- **Slow Spend Velocity Threshold**: 30% slower than usual → Flagged if Spend Velocity < Previous Spend Velocity × 0.7.

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f. Lifetime Value (LTV)

Definition:

The total expected revenue a customer will generate over their entire relationship with the business.

Logic (Description in Detail):

LTV helps determine how valuable a customer is over their entire lifecycle. A higher LTV indicates that a customer will contribute more revenue in the long term. LTV is predictive and often used for targeting high-value customers for loyalty programs or retention strategies.

Formula:

LTV = Average Purchase Value (APV) × Purchase Frequency (PF) × Customer Lifespan (CL)

Variable Values:

- **Average Purchase Value (APV)**: The average value of an order (e.g., ₹1,500).
- **Purchase Frequency (PF)**: The number of purchases per year (e.g., 4 purchases).
- **Customer Lifespan (CL)**: The expected number of years a customer will stay active (e.g., 5 years).

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g. Churn Score

Definition:

A score that estimates the likelihood of a customer churning based on their recent behavior.

Logic (Description in Detail):

Churn score is calculated using machine learning models, which consider variables like recency, frequency, and monetary factors. Customers with high churn scores are flagged as at risk of leaving.

Formula:

Churn Score = w_1 × Recency Score + w_2 × Frequency Score + w_3 × Monetary Score

Variable Values:

- **Recency Score (RS)**: Score based on the number of days since the last purchase.
- **Frequency Score (FS)**: Score based on how often a customer buys.
- **Monetary Score (MS)**: Score based on total spend.

Weights w1, w2, and w3:

These weights determine how much importance is given to each of the factors (Recency, Frequency, and Monetary) when calculating the churn score.

- **w1**: Weight for **Recency**.
  - If recency is highly important in your business model (e.g., customers who haven't purchased recently are highly likely to churn), you might assign a higher weight (e.g., 0.4 or 0.5).
- **w2**: Weight for **Frequency**.
  - If frequency is a critical factor in understanding churn (e.g., more frequent shoppers are less likely to churn), you could assign a moderate weight (e.g., 0.3).
- **w3**: Weight for **Monetary**.
  - If monetary value is the strongest indicator (e.g., high spenders are significantly less likely to churn), you could assign a higher weight to this factor as well (e.g., 0.3 or 0.4).

Example of Weighting:

For example, let's say we want to prioritize **Recency** a bit more than Frequency or Monetary (assuming recency of purchase is the most important factor for predicting churn). We could set the following values:

- w1=0.5 (high weight for recency)
- w2 = 0.3 (medium weight for frequency)
- w3=0.2 (lower weight for monetary)

Threshold:

- **High Churn Risk**: Churn Score > 70 → Flagged as high risk of churn.

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h. Dormant Bucket

Definition:

Classifies customers based on how long they've been inactive.

Logic (Description in Detail):

Customers are grouped into different dormant buckets depending on the **number of days since their last purchase**. This helps in applying different reactivation strategies.

Formula:

Dormant status = Current date – Last purchase date

Variable Values:

- **Last Purchase Date**: Date when the customer last bought something.
- **Current Date**: Today's date.

Thresholds:

- **Bucket 1**: 30–60 days → Low priority.
- **Bucket 2**: 60–90 days → Medium priority.
- **Bucket 3**: 90+ days → High priority.

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i. Win-back Eligibility

Definition:

Identifies customers who are eligible for win-back campaigns.

Logic (Description in Detail):

A customer is eligible for win-back if they have been inactive for a long period (e.g., 120+ days) but still have a significant **Lifetime Value (LTV)**.

Formula:

Win-back eligibility = Days since last purchase > 120 ∧ LTV > 2000

Variable Values:

- **LTV**: Lifetime value of the customer.
- **Days Since Last Purchase**: Number of days since the last purchase.

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j. Category Affinity

Definition:

Measures how strongly a customer is interested in specific product categories.

Logic (Description in Detail):

Tracks which product categories the customer engages with and spends the most on. This feature helps identify cross-sell opportunities.

Formula:

Category affinity = Spend in category ÷ Total spend

Variable Values:

- **Spend in Category (S)**: Total spend in a particular category (e.g., ₹5,000 on electronics).
- **Total Spend (TS)**: Total spend across all categories (e.g., ₹10,000).

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k. Offer Sensitivity

Definition:

Measures how sensitive a customer is to promotional offers.

Logic (Description in Detail):

Identifies customers who respond positively to offers (e.g., discounts, free shipping, etc.).

Formula:

Offer sensitivity = Purchases with offers / Total purchases

Variable Values:

- **Purchases with Offers (PO)**: Number of purchases made with offers (e.g., 6 out of 10).
- **Total Purchases (TP)**: Total number of purchases made by the customer (e.g., 10).

Threshold:

- **High Sensitivity**: Offer Sensitivity > 60% → Flagged as offer-sensitive.

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l. Fatigue Score

Definition:

Tracks how many campaigns a customer has been exposed to within a given period to identify if they're experiencing campaign fatigue.

Logic (Description in Detail):

Customers who are exposed to many offers in a short time are flagged for fatigue to prevent over-saturation.

Formula:

Fatigue Score = Number of Campaigns

Variable Values:

- **Number of Campaigns (NC)**: Number of campaigns the customer has been exposed to.
- **Time Period (TP)**: Time period during which the campaigns were received (e.g., 30 days).

Threshold:

- **High Fatigue**: Fatigue Score > 5 → Flagged for fatigue.

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m. Sentiment Risk Score

Definition:

Evaluates how negatively a customer feels based on interactions with support, products, or services.

Logic (Description in Detail):

A high sentiment risk score indicates that a customer is likely to churn due to dissatisfaction, and this can trigger recovery strategies.

Formula:

Sentiment Risk Score = w_1 × Negative Feedback Count + w_2 × Support Ticket Escalation Count

Variable Values:

- **Negative Feedback Count (NFC)**: Number of negative feedback entries.
- **Support Ticket Escalation Count (STEC)**: Number of escalated support tickets.

Threshold:

- **High Risk**: Sentiment Risk Score>80\text{Sentiment Risk Score} > 80Sentiment Risk Score>80 → Flag for recovery action.

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n. VIP Flag

Definition:

Flags customers who generate a significant portion of the business's revenue.

Logic (Description in Detail):

This feature is based on the **Lifetime Value (LTV)** of the customer and helps prioritize high-revenue customers for exclusive treatment.

Formula:

VIP Flag = LTV≥Top 5% LTV Threshold

Variable Values:

- **LTV**: Lifetime value of the customer.
- **Top 5% LTV Threshold**: The LTV value representing the top 5% of customers.

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o. Event Responsiveness

Definition:

Flags customers who engage with event-based campaigns (e.g., seasonal, holiday promotions).

Logic (Description in Detail):

Tracks how customers respond to seasonal or event-based offers, enabling more effective targeting during future events.

Formula:

Event responsiveness = Purchases during event ÷ Total purchases

Variable Values:

- **Purchases During Event (PDE)**: Purchases made during a seasonal or event-related campaign.
- **Total Purchases (TP)**: Total number of purchases made by the customer.

Threshold:

- **High Responsiveness**: Event Responsiveness>50%\text{Event Responsiveness} > 50\%Event Responsiveness>50% → Flag for future event campaigns.

These detailed logic descriptions, formulas, and variable values should provide a robust foundation for implementing these features in Custonomy. Let me know if you'd like to dive into more specific aspects or if you need adjustments!